ABDUL BASIT

Associate Software Engineer — AI/ML

+92 302 9230137 | abdulbaset1878@gmail.com | linkedin.com/in/abdul-basit-naeem | github.com/Abdulbaset1

SUMMARY

Computer Science graduate and software engineer specializing in AI/ML systems engineering, with hands-on experience on computer vision, NLP, and generative modeling projects end-to-end in Python and PyTorch. Strong foundation in data structures, software design, and REST APIs, paired with practical experience using AI-enabled development tools. Looking forward to get hands-on experience to real world applications.

EDUCATION

Bachelor of Science in Computer Science

FAST — National University of Computer and Emerging Sciences | Aug 2022 – Jul 2026

TECHNICAL SKILLS

- **Languages:** Python, C++, SQL
- **AI/ML & Deep Learning:** PyTorch, TensorFlow, Hugging Face Transformers, LangChain, CNNs, GANs, Transformers, Autoencoders
- **NLP:** Text Classification, Summarization, Retrieval-Augmented Generation (RAG), Seq2Seq Modeling, Machine Translation
- **Computer Vision:** Image Generation, Image-to-Image Translation, Self-Supervised Learning, Image Captioning
- **Software Engineering:** Git/GitHub, REST APIs, Linux, OOP Design, Unit Testing, CI basics, Docker (familiar)
- **Data & Tooling:** NumPy, Pandas, Scikit-learn, FAISS, OpenCV, Matplotlib, Jupyter, VS Code
- **AI-Enabled Development:** GitHub Copilot, Claude Code, Cursor

PROJECTS

RAG-Based Diagnostic Reasoning System — *Retrieval-augmented QA over clinical notes* View Repo ↗

- Designed and built a Retrieval-Augmented Generation (RAG) pipeline combining FAISS vector search with LLM-based generation via LangChain, enabling context-aware, evidence-grounded answers over unstructured clinical text.
- Architected the retrieval layer to ground generated responses in source documents, directly reducing hallucination risk — a pattern applicable to AI-enabled document review in legal, regulatory, and healthcare domains.

Document-to-Markdown VLM — *Vision-language model fine-tuned with QLoRA, deployed as an app* View Repo ↗

- Fine-tuned Qwen2-VL, a vision-language model, using QLoRA to convert document images directly into structured Markdown text.
- Built and deployed an interactive Streamlit app for uploading documents and generating/downloading Markdown output — an end-to-end, user-facing pipeline rather than a notebook-only experiment.

Doodle-to-Real Image Translation — *Conditional GAN (Pix2Pix) in PyTorch* View Repo ↗

- Built a conditional GAN (U-Net generator + PatchGAN discriminator) in PyTorch to convert hand-drawn sketches into photorealistic, automatically colorized images.
- Engineered the full training pipeline — data loading, loss functions, and evaluation — resulting in a reusable framework for supervised image-to-image translation tasks.

Domain Adaptation via CycleGAN — *Unpaired image-to-image translation* View Repo ↗

- Implemented CycleGAN in PyTorch with cycle-consistency and adversarial losses to translate images across unpaired domains (e.g., horse↔zebra, summer↔winter) without paired training data.
- Solved training instability common to GANs by tuning loss weighting and network architecture, improving output fidelity and convergence.

Self-Supervised Visual Representation Learning — *Masked Autoencoders with ViT* View Repo ↗

- Implemented Masked Autoencoders with Vision Transformers (ViT) to learn image representations by reconstructing randomly masked patches, then fine-tuned the encoder for downstream image classification.

Customer Feedback Classification Pipeline — *End-to-end NLP classification* View Repo ↗

- Built an end-to-end NLP pipeline to classify customer feedback by sentiment and topic, applying text preprocessing, feature engineering, and model evaluation to deliver actionable stakeholder insights.